# Supplementary Information

Repeatable station terms redistribute long-period response spectra across Japanese strong-motion sites

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# Supplementary Methods

## Record selection and direct periods

The public J-SHIS/NIED `sub1-v2024` flatfile contains 333,808 K-NET and KiK-net records[1, 2]. The download page defines the subset by magnitude of at least 5 and shortest fault distance no greater than 300 km. All records in the archive have JMA magnitude $M_{\mathrm{JMA}} \geq 5$; F-net $M_w$ is stored separately. Its site table codes ground-surface installations as 1 and borehole installations as 2. We retain code 1 for the residual model and response-spectrum propagation. Of 231,380 surface records, 8,675 lack finite F-net $M_w$ and 41 additional records have nonpositive AVS30. The remaining 222,664 records have positive distance, supported MF2013 source classes and finite model predictions at all eight periods.

MF2013 uses RotD100, the maximum norm of two horizontal oscillator responses[3]. RotD100 spectral acceleration is read directly at 0.1, 0.2, 0.3, 0.5, 1.0, 2.0, 3.0 and 5.0 s. Every retained record has a positive value at each period, so the final count remains 222,664 throughout. No response coordinate is copied or interpolated.

RotD50 residuals are decomposed on the identical sample as a response-component sensitivity. The RotD50 and RotD100 fields are compared station by station, and record-level RotD50/RotD100 ratios quantify the change in absolute coordinate. Zhao 2006 uses RotD50 because OpenQuake defines the model for the geometric mean of two horizontal components and the flatfile does not provide that exact response coordinate.

## Residual model boundary

The primary prediction contains the MF2013 magnitude-distance backbone and the published D1400 and AVS30 terms[4, 5]. D1400 is the depth to the 1.4 km s$^{-1}$ shear-wave velocity horizon. AVS30 is the average shear-wave velocity in the upper 30 m. The anomalous-intensity sensitivity uses the public northeastern and southwestern geographic and depth rules[6]. Incomplete plate-membership metadata restrict this term to a sensitivity calculation. The Philippine Sea Plate term is constant across stations for a qualifying event at a given period and is absorbed by the event fixed effect. The fitted global intercept is retained as a model offset; formal assessment of MF2013 calibration lies outside this design.

## Fixed effect decomposition

For residual $r_{ij}$ at station $i$ during event $j$, the additive model is

$$r_{ij} = \mu + \delta E_j + \delta S_i + \epsilon_{ij}. \tag{1}$$

Alternating grouped least squares estimates $\delta E_j$ and $\delta S_i$. Record-count-weighted event and station means are constrained to zero, so $\mu$ contains the global offset. Iteration stops when the largest parameter change falls below $10^{-10}$ $\log_{10}$ units. All 24 period-model combinations converge; the largest final change is $9.99 \times 10^{-11}$ and the largest iteration count is 1,161.

## Event group repeatability

The 1,737 usable earthquakes are assigned to five groups with a fixed random seed. The same partition is used at every period and for both ground-motion backbones. Each fold fits the two-way event-station decomposition separately to four training groups and the held-out group. Paired train and test station vectors are centred with their respective record counts. A station requires at least 15 training records and five test records. The comparison uses the train-test station correlation and the test-record-weighted root mean squared error relative to a zero station term. Each fold contains 347 or 348 test events, and every training and test decomposition converges below $10^{-10}$ $\log_{10}$ units.

## Spatial prediction

Five spatial blocks are produced by $k$-means clustering of standardised longitude and latitude. The site-only model uses shallow-velocity averages, sedimentary-depth variables, elevation, sensor depth and network. The site-and-location model adds station coordinates and two volcanic-front distances. Both histogram gradient-boosting models use fixed hyperparameters: learning rate 0.04, 400 iterations, 15 maximum leaf nodes, 35 minimum samples per leaf and L2 regularisation 0.1. Numeric variables are median-imputed within training folds; network is mode-imputed and one-hot encoded. Predictions are generated only for the held-out block and are centred to a record-weighted zero mean before spectrum propagation. A separate complexity sensitivity changes maximum leaf nodes to 7 or 31, minimum samples per leaf to 70, L2 regularisation to 1.0, or the iteration budget to 200. Each setting retains the same folds, preprocessing, weights and random seed, and no alternative is selected as a replacement for the primary model.

## MF2013 applicability sensitivity

The MF2013 regression analysis selected $M_w \geq 5.5$ ground-surface records with two horizontal components, at least five triggered stations per event and source distance below 200 km[4]. It also truncated data at the magnitude-dependent distance where the Kanno et al. equation predicted PGA below 10 cm s$^{-2}$. Our sensitivity imposes the published magnitude, distance and event-station screens on the primary flatfile sample, then repeats the residual decomposition, event holdout, spatial prediction and spectrum propagation. It contains 53,517 records from 488 earthquakes, including 772 stations with at least 20 records. The Kanno-PGA truncation is not reconstructed; the calculation tests the implemented screens and is not a reconstruction of the original regression sample.

## Alternative ground-motion-model calculation

We evaluate the Zhao et al. 2006 active-crustal, interface and intraslab equations with OpenQuake Engine 3.25.1[7, 8]. The flatfile source class selects the equation branch. Moment magnitude, hypocentral depth and crustal rake are read from the source table; documented shortest fault distance is supplied as rupture distance, and AVS30 is supplied as VS30. RotD50 is used as the closest orientation-independent coordinate to the model's geometric-mean target. The same surface observations, periods, station threshold, event folds and spatial folds are used for the MF2013 and Zhao calculations. This changes the ground-motion backbone while retaining the data and validation design.

## Transfer and event-influence tests

Cross-network tests train on K-NET and evaluate KiK-net, then reverse the direction. A common-feature model removes VS20, VS30 and network identity because these fields have different completeness in the two networks. It retains VS10, AVS30, sedimentary depths, elevation, sensor depth, coordinates and volcanic-front distances. Fixed macroregions are defined by reproducible latitude and longitude boundaries. Event influence is measured after deleting each of the ten events with the largest 3.0 s record count and after deleting all ten at every period. A record-weighted mean shift aligns each perturbed station field with the full field before comparison.

## Path and source stratification

Records are grouped into four hypocentral-bearing sectors, four shortest-fault-distance bins and three source classes. Bearing is the initial great-circle direction from the source horizontal coordinates to the station.

Stratification can disconnect the bipartite station-event graph, especially within 50 km. Each graph component is solved separately by sparse least squares. Stations require ten records within a stratum, retained components require ten supported stations, and every component is aligned to the full station field by its record-weighted mean difference. Across all 88 period-stratum fits, the maximum relative normal-equation residual is $1.90 \times 10^{-9}$ and every solver returns an accepted convergence code.

Five random, disjoint event-half comparisons at 1.0, 2.0 and 3.0 s measure sampling reliability within each stratum. A station requires five records in each half. Both half-fields are component-aligned for storage and for constructing equal-stratum targets. The reported split-half correlation removes the weighted mean from every intersecting component pair before pooling stations, making the statistic invariant to unidentifiable component offsets. Equal-stratum sensitivity targets are formed by averaging the aligned station terms across at least two available strata, with equal weight per stratum. The primary spatial-validation procedure is then repeated for azimuth-, distance- and source-class-balanced targets; prediction errors retain station record-count weights.

## Empirical prediction intervals

For each held spatial block, signed prediction errors from the other four blocks provide record-weighted 5th and 95th percentiles. These quantiles are added to the held-block point prediction. Coverage is computed without using the held-block station terms for interval calibration. Lower and upper station terms are converted to multipliers and propagated to the matched response spectra. The resulting interval describes station-model prediction error under this spatial split. Source, ground-motion-model and J-SHIS logic-tree uncertainties are outside its scope.

## Reference condition conversion

J-SHIS response-spectrum maps provide ordinates on engineering bedrock with $V_S = 400$ m s$^{-1}$[9, 10]. The station-surface factor replaces the MF2013 shallow-soil term for 400 m s$^{-1}$ with the term evaluated at the station AVS30:

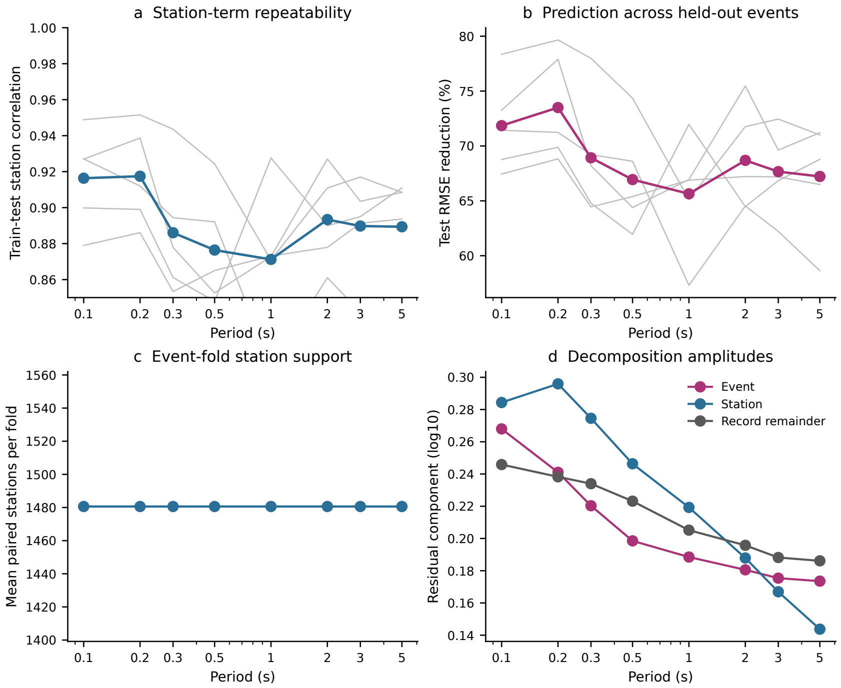
$$F_{\mathrm{surf},i}(T) = 10^{G_s(AVS30_i,T)-G_s(400,T)}. \tag{2}$$

The cross-validated station factor $10^{\widehat{\delta S}_i(T)}$ is applied after this conversion. All eight period coordinates and four 50-year exceedance probabilities are read from the official map files. The calculation is restricted to matched station cells; national-grid interpolation lies outside this analysis.

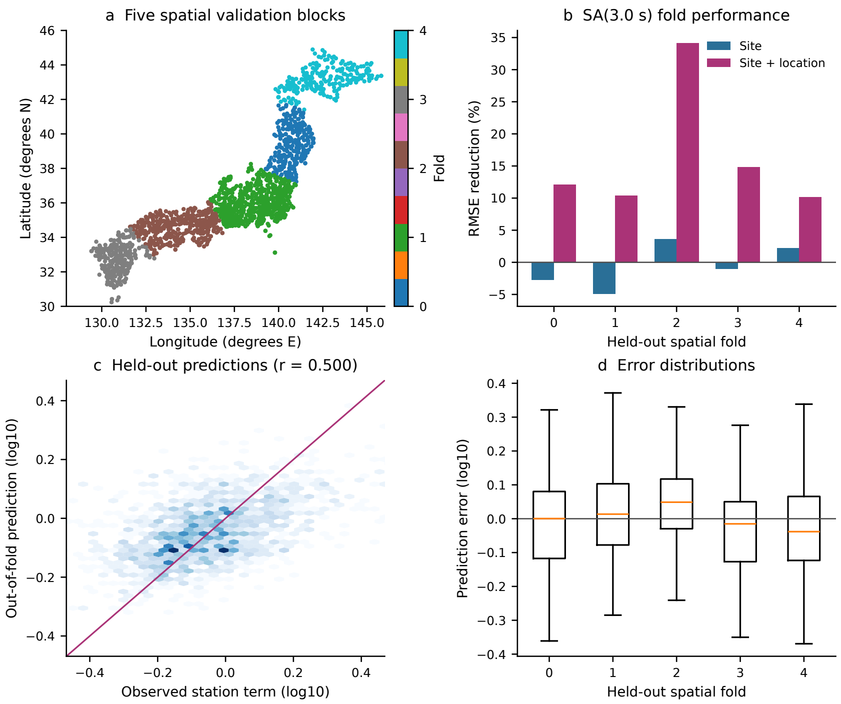
## KiK-net waveform check

The waveform check uses 460 paired surface and downhole records from two KiK-net earthquakes[11]. The first set contains 288 converted miniSEED pairs and the second contains 172 original NIED ASCII pairs converted with the scale factors in their headers. Each trace is demeaned and detrended. Horizontal peak amplitude is calculated from the two horizontal components. Surface-to-downhole Fourier amplitude ratios are evaluated from 0.2 to 20 Hz. This check is independent of the station residual model.

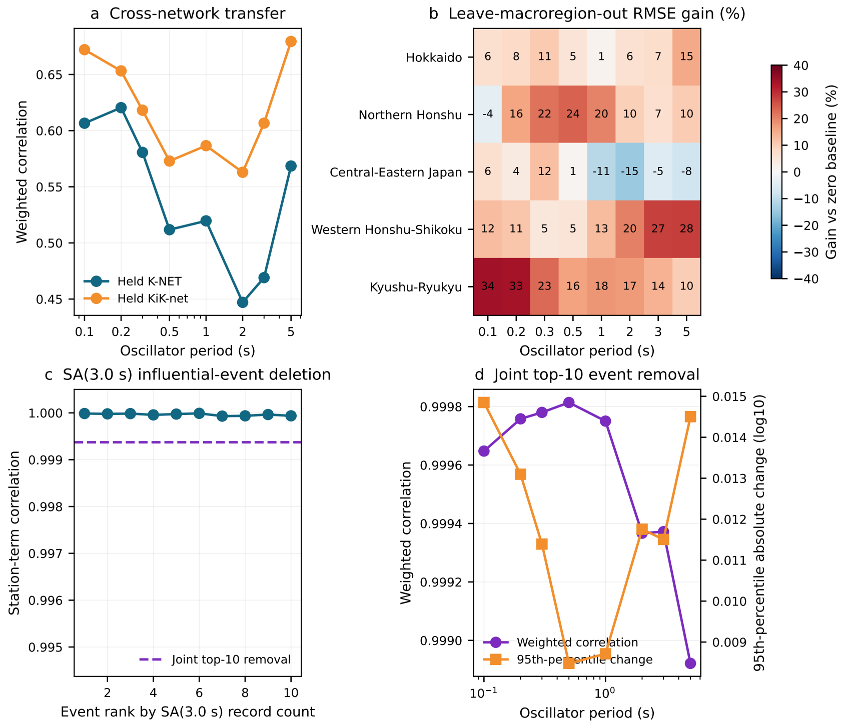
# Supplementary Figures



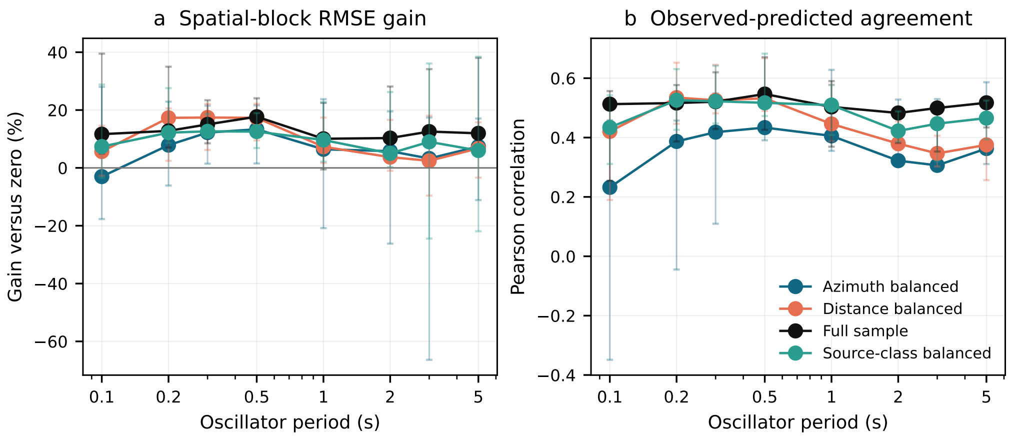

**Supplementary Figure 1. Station term repeatability across held-out events.** **a**, train-test station correlations for five event folds; grey lines show individual folds and the blue line shows their mean. **b**, test root mean squared error reduction relative to a zero station term. **c**, mean number of stations meeting train and test record thresholds. **d**, event, station and record-remainder amplitudes from the primary MF2013 site residual decomposition.



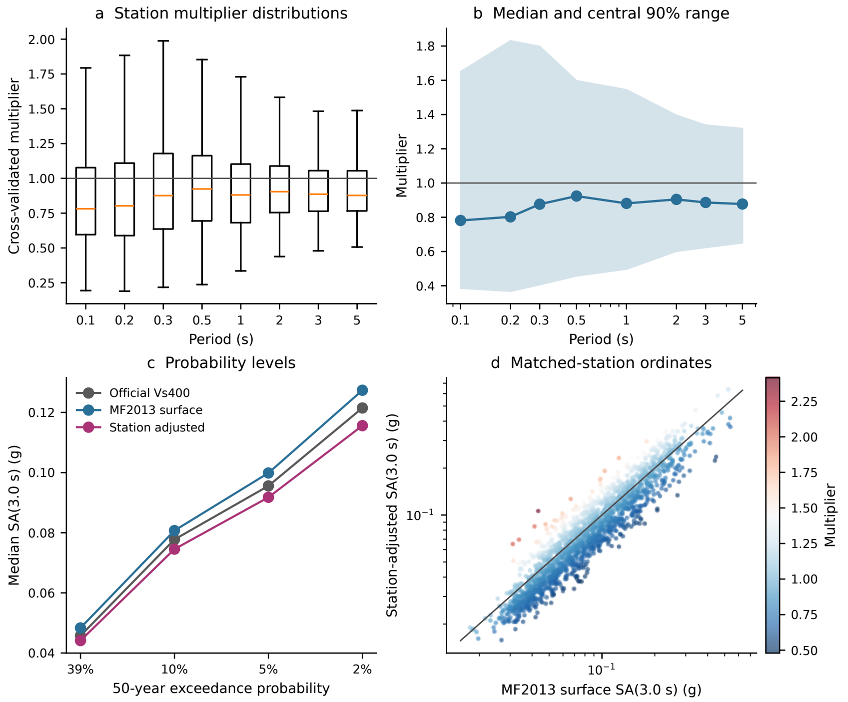

**Supplementary Figure 2. Spatial validation of SA(3.0 s) station prediction.** **a**, five station blocks used for cross-validation. **b**, root mean squared error reduction in each held-out block for the site-only and site-plus-location models. **c**, observed event-adjusted station terms and out-of-fold site-plus-location predictions. **d**, prediction-error distributions by spatial fold; boxes show interquartile ranges and whiskers extend to 1.5 times the interquartile range.



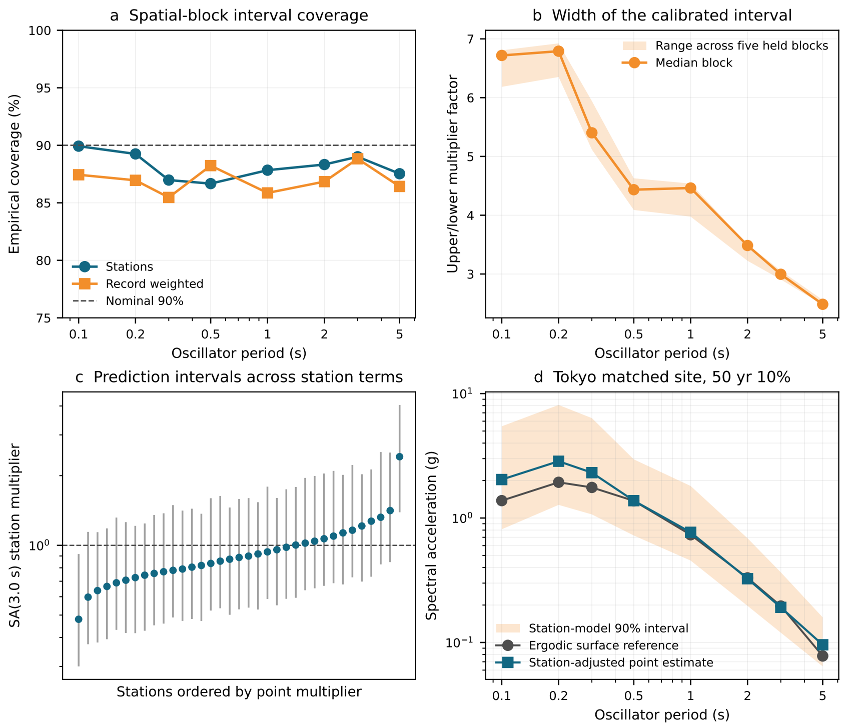

**Supplementary Figure 3. Network, geographic and event-influence stress tests.** **a**, common-feature transfer between K-NET and KiK-net. **b**, common-feature root mean squared error gain when each fixed macroregion is held out. **c**, SA(3.0 s) station-term correlation after deleting each of the ten highest-count events; the dashed line shows their joint deletion. **d**, station-term correlation and weighted 95th-percentile absolute change after jointly deleting the ten events at every period.



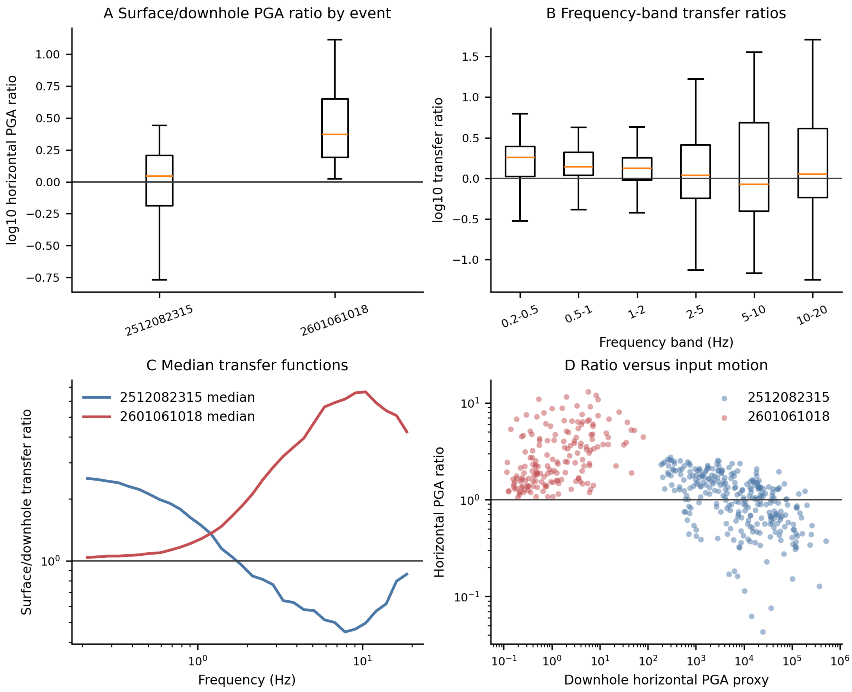

**Supplementary Figure 4. Prediction of equal-stratum station fields.** **a**, overall spatial-block root mean squared error gain relative to zero. **b**, observed-predicted correlation. Vertical bars span the five held-block results; circles show pooled results. The full-sample target is included for reference.




**Supplementary Figure 5. Matched-station response-spectrum sensitivity.** **a**, cross-validated station multiplier distributions at the eight direct periods. **b**, median and 5th–95th percentile multiplier range. **c**, median SA(3.0 s) ordinates at four 50-year exceedance probabilities for the official $V_S = 400$ m s$^{-1}$ map, the MF2013 surface projection and the station-adjusted surface projection. **d**, paired MF2013 surface and station-adjusted SA(3.0 s) values at the 50-year 10% level; colour denotes the station multiplier.




**Supplementary Figure 6. Spatial-block station-model uncertainty.** **a**, empirical coverage of the nominal 90% interval by period. **b**, median interval factor span and its range across the five held blocks. **c**, point multipliers and empirical intervals for 35 ordered SA(3.0 s) stations. **d**, surface-reference and station-adjusted spectra with the station-model interval at the Tokyo-nearest matched station. The interval excludes source, backbone and J-SHIS logic-tree uncertainty.




**Supplementary Figure 7. KiK-net surface and downhole waveform comparison.** **a**, horizontal surface-to-downhole peak acceleration ratios by earthquake. **b**, frequency-band Fourier amplitude ratios. **c**, median transfer functions for the two event sets. **d**, horizontal peak acceleration ratio against the downhole input-motion proxy. Ratios are shown in $\log_{10}$ units in panels **a** and **b**.

# Supplementary Tables

**Supplementary Table 1.** Primary residual decomposition

| Period (s) | Records | Global offset | Event SD | Station SD | Remainder RMSE |
|---|---|---|---|---|---|
| | | | $\log_{10}$ units | | |
| 0.1 | 222,664 | 0.284 | 0.268 | 0.284 | 0.246 |
| 0.2 | 222,664 | 0.215 | 0.241 | 0.296 | 0.238 |
| 0.3 | 222,664 | 0.172 | 0.220 | 0.275 | 0.234 |
| 0.5 | 222,664 | 0.106 | 0.199 | 0.246 | 0.223 |
| 1.0 | 222,664 | 0.017 | 0.188 | 0.219 | 0.205 |
| 2.0 | 222,664 | -0.039 | 0.181 | 0.188 | 0.196 |
| 3.0 | 222,664 | -0.085 | 0.175 | 0.167 | 0.188 |
| 5.0 | 222,664 | -0.134 | 0.174 | 0.144 | 0.186 |

**Supplementary Table 2.** Event group station repeatability

| Period (s) | Train-test correlation | Test RMSE reduction (%) | Mean paired stations | Mean test events |
|---|---|---|---|---|
| 0.1 | 0.916 | 71.9 | 1,481 | 347.4 |
| 0.2 | 0.917 | 73.5 | 1,481 | 347.4 |
| 0.3 | 0.886 | 68.9 | 1,481 | 347.4 |
| 0.5 | 0.876 | 66.9 | 1,481 | 347.4 |
| 1.0 | 0.871 | 65.6 | 1,481 | 347.4 |
| 2.0 | 0.893 | 68.7 | 1,481 | 347.4 |
| 3.0 | 0.890 | 67.7 | 1,481 | 347.4 |
| 5.0 | 0.889 | 67.2 | 1,481 | 347.4 |

**Supplementary Table 3.** Zhao 2006 ground-motion-model sensitivity

| Period (s) | MF2013–Zhao correlation | Median absolute difference ($\log_{10}$) | Zhao event correlation | Zhao spatial RMSE gain (%) |
|---|---|---|---|---|
| 0.1 | 0.953 | 0.064 | 0.940 | 16.9 |
| 0.2 | 0.921 | 0.093 | 0.941 | 20.8 |
| 0.3 | 0.919 | 0.081 | 0.913 | 18.9 |
| 0.5 | 0.923 | 0.073 | 0.909 | 20.8 |
| 1.0 | 0.889 | 0.094 | 0.915 | 22.9 |
| 2.0 | 0.801 | 0.144 | 0.948 | 32.1 |
| 3.0 | 0.770 | 0.151 | 0.950 | 37.7 |
| 5.0 | 0.697 | 0.125 | 0.946 | 39.2 |

**Supplementary Table 4.** Spatially held-out station prediction

| Period (s) | Site-only model RMSE gain (%) | Site and location RMSE gain (%) | Observed-predicted correlation |
|---|---|---|---|
| 0.1 | -5.2 | 11.7 | 0.513 |
| 0.2 | 4.3 | 12.8 | 0.516 |
| 0.3 | 10.3 | 15.0 | 0.521 |
| 0.5 | 9.0 | 17.7 | 0.547 |
| 1.0 | 4.6 | 10.1 | 0.503 |
| 2.0 | 0.0 | 10.3 | 0.483 |
| 3.0 | -2.3 | 12.6 | 0.500 |
| 5.0 | -3.5 | 12.0 | 0.517 |

**Supplementary Table 5.** SA(3.0 s) network and fixed-macroregion transfer

| Held group | Full-model RMSE gain (%) | Common-feature RMSE gain (%) | Common-feature correlation |
|---|---|---|---|
| K-NET | 4.1 | 8.7 | 0.469 |
| KiK-net | 21.2 | 20.5 | 0.607 |
| Hokkaido | 7.2 | 6.9 | 0.284 |
| Northern Honshu | 5.4 | 6.8 | 0.304 |
| Central-Eastern Japan | 6.2 | -5.2 | 0.402 |
| Western Honshu–Shikoku | 28.9 | 27.5 | 0.386 |
| Kyushu–Ryukyu | 13.4 | 13.8 | 0.372 |

**Supplementary Table 6.** Joint deletion of the ten highest-count events

| Period (s) | Station-term correlation | Weighted 95th-percentile absolute change ($\log_{10}$) |
|---|---|---|
| 0.1 | 0.99965 | 0.0148 |
| 0.2 | 0.99976 | 0.0131 |
| 0.3 | 0.99978 | 0.0114 |
| 0.5 | 0.99981 | 0.0085 |
| 1.0 | 0.99975 | 0.0087 |
| 2.0 | 0.99937 | 0.0118 |
| 3.0 | 0.99937 | 0.0115 |
| 5.0 | 0.99892 | 0.0145 |

**Supplementary Table 7.** SA(3.0 s) path and source stratification

| Stratification | Stratum | Paired stations | Full-field correlation | Within-component split-half correlation | 95th percentile difference |
|---|---|---|---|---|---|
| Hypocentral azimuth | 000–090° | 938 | 0.280 | 0.884 | 0.474 |
| Hypocentral azimuth | 090–180° | 448 | 0.678 | 0.764 | 0.236 |
| Hypocentral azimuth | 180–270° | 1,089 | 0.777 | 0.969 | 0.317 |
| Hypocentral azimuth | 270–360° | 1,345 | 0.886 | 0.945 | 0.168 |
| Fault distance | 0–50 km | 138 | 0.644 | 0.631 | 0.289 |
| Fault distance | 50–100 km | 680 | 0.496 | 0.793 | 0.427 |
| Fault distance | 100–200 km | 1,515 | 0.974 | 0.951 | 0.080 |
| Fault distance | 200–300 km | 1,213 | 0.963 | 0.968 | 0.092 |
| Source type | Crustal | 1,494 | 0.942 | 0.905 | 0.120 |
| Source type | Interplate | 960 | 0.934 | 0.958 | 0.073 |
| Source type | Intraplate | 1,379 | 0.924 | 0.902 | 0.131 |

**Supplementary Table 8.** Equal-stratum spatial prediction sensitivity

| Target field | Period (s) | Stations | Overall RMSE reduction (%) | Overall correlation | Minimum fold reduction (%) |
|---|---|---|---|---|---|
| Azimuth balanced | 1.0 | 1,220 | 6.4 | 0.406 | -20.8 |
| Azimuth balanced | 2.0 | 1,220 | 5.8 | 0.322 | -26.2 |
| Azimuth balanced | 3.0 | 1,220 | 3.2 | 0.306 | -66.4 |
| Distance balanced | 1.0 | 1,285 | 7.3 | 0.446 | 1.7 |
| Distance balanced | 2.0 | 1,285 | 3.7 | 0.379 | -1.0 |
| Distance balanced | 3.0 | 1,285 | 2.5 | 0.347 | -9.6 |
| Source balanced | 1.0 | 1,369 | 9.6 | 0.509 | 2.2 |
| Source balanced | 2.0 | 1,369 | 5.0 | 0.422 | 0.3 |
| Source balanced | 3.0 | 1,369 | 9.0 | 0.447 | -24.5 |

**Supplementary Table 9.** Primary full-sample matched-station 50-year 10 percent response spectra

| Period | Station multiplier | | | Official $V_S400$ | MF2013 surface | Station adjusted |
|---|---|---|---|---|---|---|
| (s) | 5th | Median | 95th | median spectral acceleration (g) | | |
| 0.1 | 0.386 | 0.781 | 1.650 | 0.946 | 0.978 | 0.817 |
| 0.2 | 0.368 | 0.802 | 1.830 | 1.068 | 1.154 | 0.936 |
| 0.3 | 0.407 | 0.876 | 1.797 | 0.914 | 1.022 | 0.891 |
| 0.5 | 0.458 | 0.924 | 1.597 | 0.652 | 0.737 | 0.668 |
| 1.0 | 0.498 | 0.881 | 1.544 | 0.315 | 0.341 | 0.308 |
| 2.0 | 0.602 | 0.904 | 1.395 | 0.132 | 0.140 | 0.130 |
| 3.0 | 0.624 | 0.886 | 1.338 | 0.078 | 0.081 | 0.075 |
| 5.0 | 0.651 | 0.877 | 1.318 | 0.030 | 0.031 | 0.028 |

**Supplementary Table 10.** Empirical station-model prediction intervals

| Period (s) | Station coverage (%) | Record-weighted coverage (%) | Median upper/lower multiplier factor |
|---|---|---|---|
| 0.1 | 89.9 | 87.4 | 6.72 |
| 0.2 | 89.3 | 87.0 | 6.79 |
| 0.3 | 87.0 | 85.5 | 5.40 |
| 0.5 | 86.7 | 88.2 | 4.43 |
| 1.0 | 87.8 | 85.9 | 4.32 |
| 2.0 | 88.3 | 86.8 | 3.49 |
| 3.0 | 89.0 | 88.8 | 3.00 |
| 5.0 | 87.5 | 86.4 | 2.48 |

**Supplementary Table 11.** City-nearest SA(3.0 s) station cases

| City | Station | Distance (km) | Multiplier | Change (%) | Surface SA | Adjusted SA |
|---|---|---|---|---|---|---|
| | | | | | (g) | |
| Tokyo | TKY007 | 2.4 | 0.975 | -2.5 | 0.196 | 0.192 |
| Osaka | OSK005 | 3.6 | 0.885 | -11.5 | 0.223 | 0.197 |
| Sapporo | HKD180 | 8.6 | 1.312 | +31.2 | 0.124 | 0.163 |
| Sendai | MYG013 | 5.0 | 0.890 | -11.0 | 0.131 | 0.116 |
| Fukuoka | FKOH03 | 14.1 | 0.748 | -25.2 | 0.035 | 0.026 |

**Supplementary Table 12.** Sequential record selection

| Stage | Records | Excluded | Earthquakes | Stations |
|---|---|---|---|---|
| Public `sub1-v2024` archive | 333,808 | 0 | 1,840 | 2,581 |
| Ground-surface installation | 231,380 | 102,428 | 1,840 | 1,882 |
| Finite F-net $M_w$ | 222,705 | 8,675 | 1,737 | 1,881 |
| Supported source class and positive distance | 222,705 | 0 | 1,737 | 1,881 |
| Positive AVS30 | 222,664 | 41 | 1,737 | 1,880 |
| Finite RotD100 and MF2013 predictions at all periods | 222,664 | 0 | 1,737 | 1,880 |

**Supplementary Table 13.** Response-component sensitivity

| Period (s) | Station-field correlation | Weighted difference RMSE | 95th percentile absolute difference | Median record RotD50/RotD100 |
|---|---|---|---|---|
| 0.1 | 0.9995 | 0.0076 | 0.0179 | 0.853 |
| 0.2 | 0.9995 | 0.0081 | 0.0183 | 0.839 |
| 0.3 | 0.9994 | 0.0079 | 0.0176 | 0.834 |
| 0.5 | 0.9993 | 0.0073 | 0.0177 | 0.831 |
| 1.0 | 0.9990 | 0.0077 | 0.0197 | 0.826 |
| 2.0 | 0.9988 | 0.0074 | 0.0194 | 0.820 |
| 3.0 | 0.9986 | 0.0071 | 0.0180 | 0.819 |
| 5.0 | 0.9983 | 0.0064 | 0.0177 | 0.820 |

**Supplementary Table 14.** SA(3.0 s) spatial-model complexity sensitivity

| Setting | Changed parameter | RMSE gain (%) | Correlation |
|---|---|---|---|
| Primary | None | 12.56 | 0.500 |
| Fewer leaves | Maximum leaf nodes = 7 | 13.98 | 0.512 |
| More leaves | Maximum leaf nodes = 31 | 10.45 | 0.467 |
| Larger minimum leaf | Minimum samples per leaf = 70 | 10.71 | 0.483 |
| Stronger regularisation | L2 regularisation = 1.0 | 12.29 | 0.496 |
| Shorter iteration budget | Maximum iterations = 200 | 12.85 | 0.497 |

**Supplementary Table 15.** MF2013 magnitude, distance and event-station sensitivity

| Period (s) | Station-field correlation | 95th-percentile difference ($\log_{10}$) | Event-group correlation | Spatial RMSE gain (%) | OOF-prediction correlation | Restricted multiplier 5th–95th percentile |
|---|---|---|---|---|---|---|
| 0.1 | 0.977 | 0.120 | 0.940 | -2.6 | 0.780 | 0.443–1.936 |
| 0.2 | 0.981 | 0.117 | 0.946 | 12.4 | 0.849 | 0.419–1.868 |
| 0.3 | 0.981 | 0.112 | 0.939 | 17.5 | 0.778 | 0.471–1.956 |
| 0.5 | 0.981 | 0.097 | 0.931 | 20.2 | 0.819 | 0.493–1.773 |
| 1.0 | 0.976 | 0.098 | 0.927 | 12.1 | 0.662 | 0.543–1.656 |
| 2.0 | 0.970 | 0.089 | 0.897 | 10.6 | 0.664 | 0.612–1.561 |
| 3.0 | 0.959 | 0.083 | 0.895 | 9.9 | 0.770 | 0.649–1.445 |
| 5.0 | 0.921 | 0.103 | 0.834 | 9.2 | 0.753 | 0.674–1.381 |

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